5. Identify the like terms : $12x^3y^2z,\ -y^3x^2z,\ 4z^3y^2x,\ 6x^3z^2y,\ -5y^3x^2z$

6. Add and find the degree of the following expressions.

   (i) $(9x+3y)$ and $(10x-9y)$ (ii) $\left(k^2-25k+46\right)$ and $\left(23-2k^2+21k\right)$

   (iii) $\left(3m^2n+4pq^2\right)$ and $(5nm^2-2q^2p)$

7. Simplify and find the degree of the following expressions.

   (i) $10x^2-3xy+9y^2-\left(3x^2-6xy-3y^2\right)$ (ii) $9a^4-6a^3-6a^4-3a^2+7a^3+5a^2$

   (iii) $4x^2-3x-\left[8x-\left(5x^2-8\right)\right]$

Objective type questions

8. $3p^2-5pq+2q^2+6pq-q^2+pq$ is a

   (i) Monomial (ii) Binomial (iii) Trinomial (iv) Quadrinomial

9. The degree of $6x^7-7x^3+4$ is

   (i) 7 (ii) 3 (iii) 6 (iv) 4

10. If $p(x)$ and $q(x)$ are two expressions of degree 3, then the degree of $p(x)+q(x)$ is

    (i) 6 (ii) 0 (iii) 3 (iv) Undefined

Exercise 3.4

Miscellaneous Practice problems

1. $6^2\times 6^m = 6^5$, find the value of 'm'.

2. Find the unit digit of $124^{128}\times 126^{124}$.

3. Find the unit digit of the numeric expression: $16^{23}+71^{48}+59^{61}$

4. Find the value of $\dfrac{(-1)^6\times(-1)^7\times(-1)^8}{(-1)^3\times(-1)^5}$.

5. Identify the degree of the expression, $2a^3bc+3a^3b+3a^3c-2a^2b^2c^2$

6. If $p=-2,\ q=1$ and $r=3$, find the value of $3p^2q^2r$.

Challenge Problems

7. **LEADERS** is a WhatsApp group with 256 members. Every one of its member is an admin for their own WhatsApp group with 256 distinct members. When a message is posted in LEADERS and everybody forwards the same to their own group, then how many members in total will receive that message?

8. Find x such that $3^{x+2}=3^x+216$.

9. If $X=5x^2+7x+8$ and $Y=4x^2-7x+3$, then find the degree of $X+Y$.

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